Finite State Machine Practice Problems

Note: Unless otherwise stated, assume a synchronous sequential machine with one input X. Outputs are Moore or Mealy as specified.

State Reduction

Problem 1: Moore Machine Reduction (Partition Refinement)

A sequential circuit is described by the following Moore state table:

Present State	X=0	X=1	Output
A	B	C	0
B	D	E	0
C	D	E	0
D	D	F	1
E	D	F	1
F	F	F	1

Tasks:

- Identify equivalent states using partition refinement and combine equivalent states.
- Show all partition refinement steps clearly.
- Write the final reduced state table.
- Draw the reduced state diagram after reduction.
- State the number of states before and after reduction.

Problem 2: Mealy Machine Reduction (Implication Table)

A Mealy machine is defined by the following state table:

Present State	X=0	X=1
A	B/0	C/1
B	D/0	E/1
C	D/0	E/1
D	D/0	F/1
E	D/0	F/1
F	F/1	F/1

Tasks:

- Construct the implication table and identify equivalent state pairs.
- Combine equivalent states and obtain the minimized state table.
- Draw the reduced state diagram after reduction.
- State the number of states before and after reduction.

Problem 3: Partition Refinement with Multiple Groups

A Moore machine is given below:

Present State	X=0	X=1	Output
A	B	C	0
B	D	E	0
C	F	G	0
D	D	E	1
E	D	E	1
F	F	G	1
G	F	G	1

Tasks:

- Use partition refinement to identify equivalent states and combine equivalent states.
- Show all intermediate partitions.
- Write the final reduced state table.
- Draw the reduced state diagram after reduction.
- State the number of states before and after reduction.

Problem 4: Large FSM Reduction

Consider the following Moore state machine:

Present State	X=0	X=1	Output
A	B	C	0
B	D	E	0
C	F	G	0
D	D	H	1
E	D	H	1
F	F	H	1
G	F	H	1
H	H	H	1

Tasks:

- Use partition refinement to minimize the machine.
- Show all partition refinement steps clearly.
- Write the final reduced state table.
- Draw the reduced state diagram after reduction.
- State the number of states before and after reduction.

State Assignment

Problem 5: Binary State Assignment and Implementation

A sequential machine has five states: A, B, C, D, and E.

Transitions:

- A $\rightarrow$ B (X=0), C (X=1)
- B $\rightarrow$ D (X=0), E (X=1)
- C $\rightarrow$ D (X=0), E (X=1)
- D $\rightarrow$ D (X=0), E (X=1)
- E $\rightarrow$ E (for both inputs)

Tasks:

- Use binary state assignment to encode the states.
- Implement the circuit using D flip-flops.
- Derive the next-state equations (D inputs).
- Derive the output equation, if required.

Problem 6: One-Hot State Assignment

Present State	X=0	X=1
A	B	C
B	D	A
C	D	A
D	D	D

Tasks:

- Use one-hot encoding for state assignment.
- Derive the next-state equations (D flip-flop inputs).
- Draw the logic diagram for the implementation.

Problem 7: Optimal State Assignment (Logic Minimization)

A sequential system has five states: A, B, C, D, and E.

Transitions:

- A $\rightarrow$ B, C
- B $\rightarrow$ D, E
- C $\rightarrow$ D, E
- D $\rightarrow$ A
- E $\rightarrow$ A

Tasks:

- Choose a state assignment that minimizes the next-state logic.
- Justify your assignment based on similarity of transitions.
- Derive simplified next-state equations using D flip-flops.

Problem 8: State Assignment with Output Optimization

State	Output
A	0
B	0
C	1
D	1
E	1

Present State	X=0	X=1
A	B	C
B	D	E
C	D	E
D	D	E
E	E	E

Tasks:

- Choose a state assignment that minimizes the output logic.
- Derive the output Boolean expression.
- Derive the next-state equations using D flip-flops.
- Compare your assignment with a simple binary assignment.